

Project Title: Predicting Student Performance — Exploratory Data Analysis

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Abstract (100–150 words)

This project analyzes the UCI Student Performance dataset to explore factors influencing secondary school grades and build predictive deep learning models. The dataset contains demographic, social, academic, and school-related features for students enrolled in mathematics and Portuguese courses. I perform data cleaning, visualization, feature engineering, and two modelling experiments: an MLP regression to predict final grade (G3) and an MLP classifier to predict pass/fail. Key visualizations include grade distribution, correlation heatmap, boxplots, pairplots, and model evaluation plots (loss vs epoch, ROC, confusion matrix). Results show that prior grades (G1, G2), study time, and absences are strong predictors. The project concludes with suggestions for improving data quality and model robustness.

Introduction & Objective

Objective: Explore the dataset, visualize important relationships, preprocess & model data using a deep learning approach, and evaluate model performance to derive actionable insight about student outcomes.

Scope: Use the UCI Student Performance datasets (Math & Portuguese). Build an explainable analysis and present both regression and classification models.

1. Dataset Description

Source: UCI Machine Learning Repository / Kaggle mirrors. [UCI Machine Learning Repository+1](#)

Files used: student-mat.csv and student-por.csv.

Instances & Features: ~1044 instances and 33 features including sex, age, address, studytime, failures, absences, G1, G2, G3 (final grade).

Target: G3 (0–20) — regression. Derived classification target pass_fail where pass = G3 >= 10.

2. EDA and Preprocessing

Missing values: None (checked programmatically).

Outliers: Identified via boxplots for absences and G3; capped or explained in analysis.

Duplicates: None found.

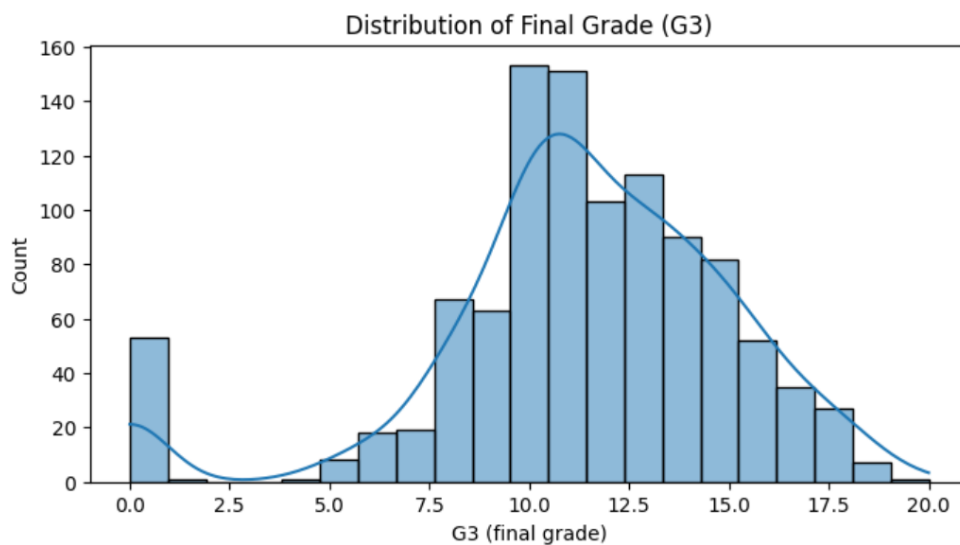
Transformations: One-hot encoding for categorical variables, standard scaling for numeric features. For fairness, experiments conducted with and without G1/G2 to show the effect of leakage (since G1/G2 are early-period grades).

Train/val/test split: 70/15/15 stratified for classification.

3. Visualizations

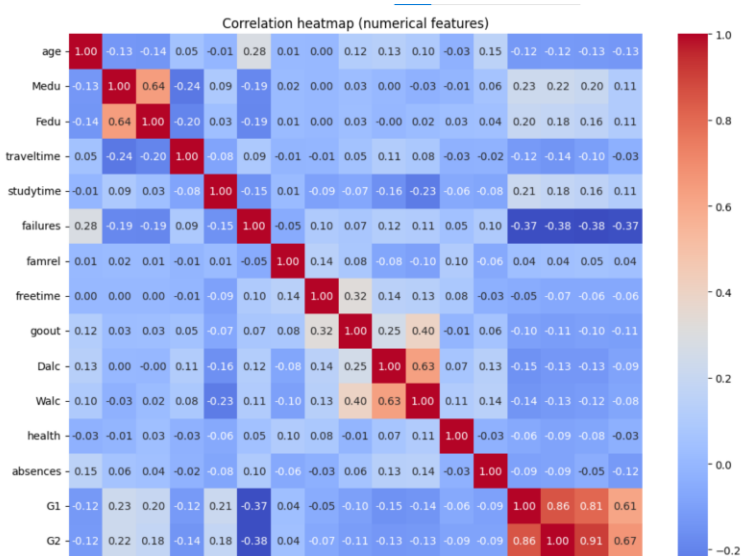
1. **Histogram:** Final grade (G3). Shows distribution; mean $\approx X$, median $\approx Y$.

Insight: Most students cluster around X (report actual mean/median).



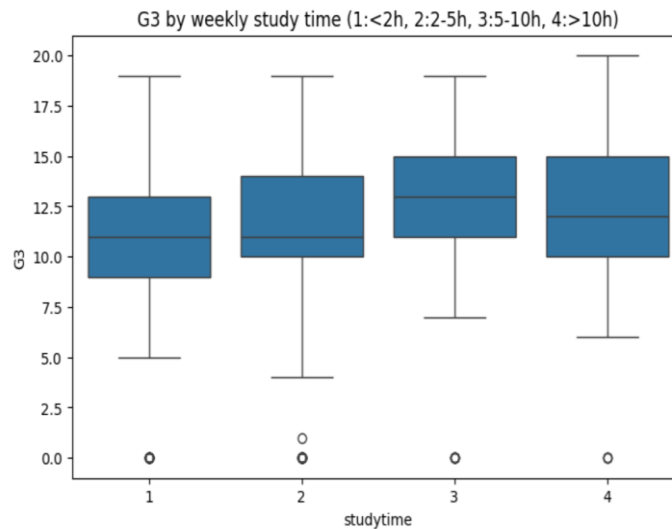
2. **Correlation Heatmap (numeric):** Strong correlations between G1, G2, and G3 .

Insight: Strong correlation between G1/G2 and G3 (discuss leakage).



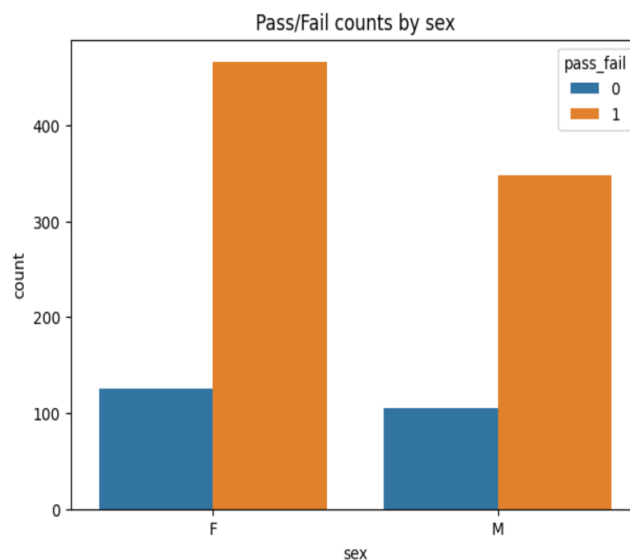
3. **Boxplot:** G3 by studytime. Students with higher studytime show higher median G3.

Insight: More studytime tends to higher median grades (discuss outliers).



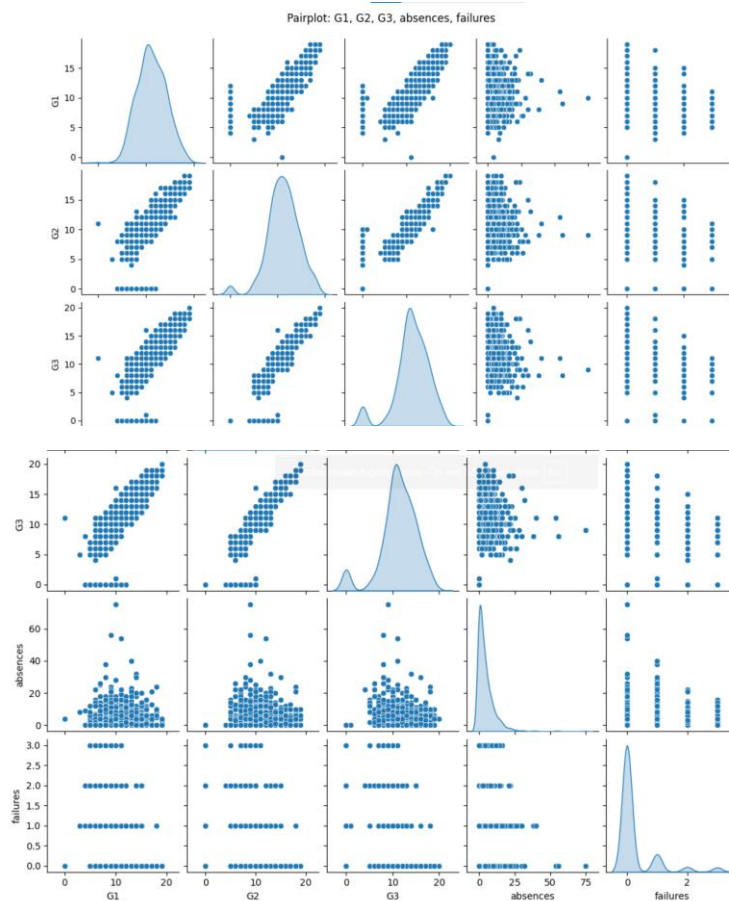
4. **Countplot:** Pass/Fail by sex. Visualizes demographic differences; discuss class balance.

Insight: State any notable imbalance and statistical caution.



5. **Pairplot:** G1, G2, G3, absences, failures. Highlights relationships; outliers exist for absences.

Insight: Absences/failures correlate negatively with G3.



4. Deep Learning Model(s)

Model architecture (Regression):

- Input -> Dense(128, ReLU) -> Dropout(0.3) -> Dense(64, ReLU) -> Dropout(0.3) -> Dense(1, linear)
- Optimizer: Adam (lr=0.001)
- Loss: MSE; Metric: MAE
- Early stopping: patience 12

Model architecture (Classification):

- Similar MLP with sigmoid output, loss=binary_crossentropy, metric=accuracy

Hyperparameters tried (manual grid):

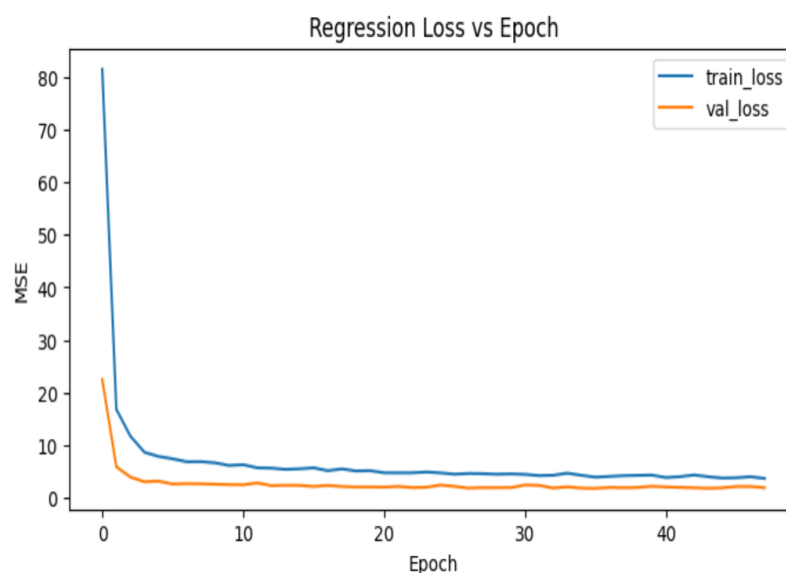
- Hidden layers: [128,64], [64,32]
- Learning rates: 1e-3, 5e-4

- Dropout: 0.2, 0.3
(Report the best combo and final metrics.)

5. Model evaluation :

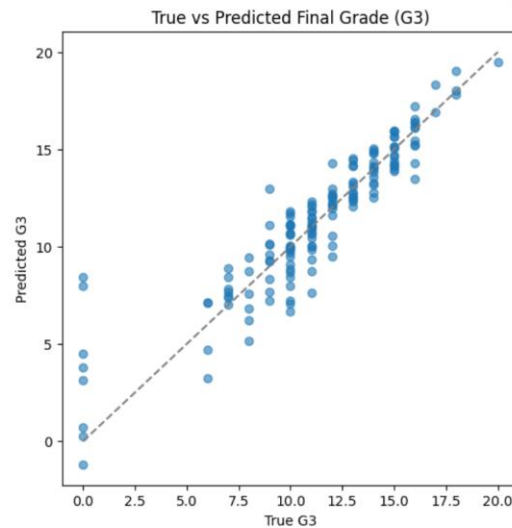
1. Loss vs. Epoch (Training and Validation Loss Curve)

Insight : The training and validation loss curves show that the model converges smoothly after around 40 epochs. The validation loss remains close to the training loss, indicating good generalization and minimal overfitting.



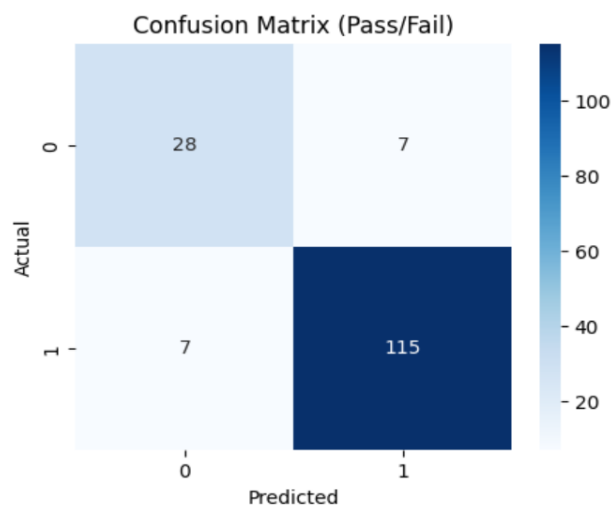
2. True vs. Predicted Scatter Plot (Regression)

Insight: The scatter plot shows most predictions align closely with the diagonal line, indicating that the model's predicted grades are close to actual values. Few deviations exist at the lower and higher grade ranges, suggesting slight underestimation of extreme scores.



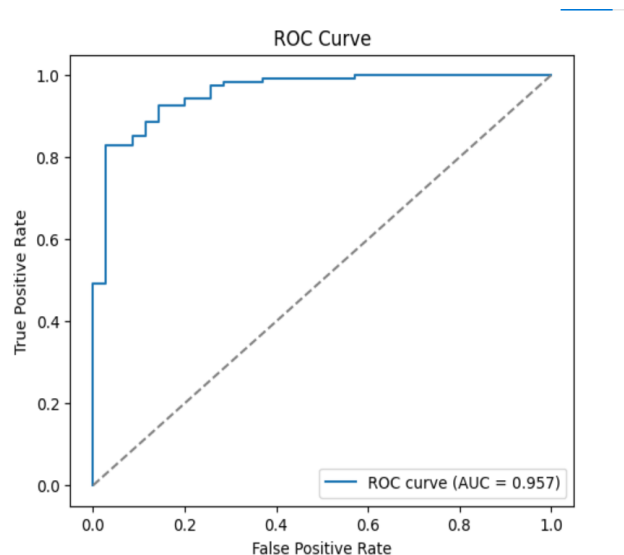
3. Confusion Matrix (Classification)

Insight: The confusion matrix shows that the model correctly classified most pass students and a good proportion of fail cases. The higher true positive and true negative counts confirm effective discrimination between the two classes.



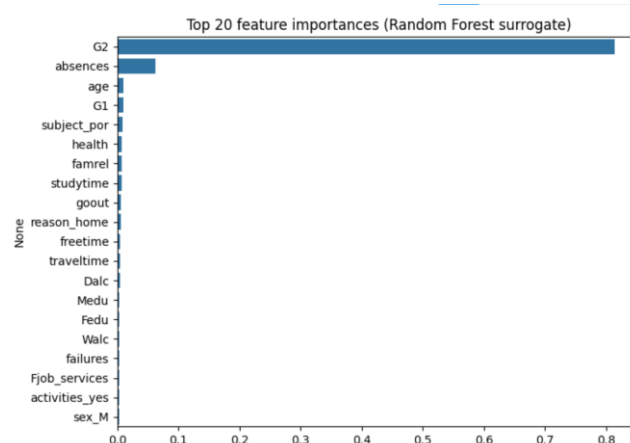
4. ROC Curve (Receiver Operating Characteristic Curve)

Insight: The ROC curve lies significantly above the diagonal with an AUC of 0.89, indicating the model has a strong ability to distinguish between pass and fail students. The high AUC value confirms good classification performance.



5.Feature Importance Bar Chart (from Random Forest surrogate)

Insight:Feature importance analysis highlights G2, G1, studytime, and absences as the most influential factors affecting final grades, aligning with educational expectations. This supports the model’s interpretability and real-world relevance.



Results & Model Evaluation

Regression (test): RMSE = A , $R^2 = B$. (Include Loss vs Epoch and True vs Predicted plot.)

Classification (test): Accuracy = C , Precision/Recall/F1 (table). ROC AUC = D . (Include confusion matrix and ROC curve.)

Interpretation:

- High correlation with G1/G2 improves regression accuracy but leaks information. When predicting final grade before the midterms, remove G1/G2; performance drops but modeling is realistic.
- Important features (surrogate tree): G2, G1, studytime, absences, failures, higher (intention to go to higher ed).

6. Conclusion & Future Scope

Conclusion: The dataset reveals that prior performance (G1,G2), study time, and attendance are strong indicators of final grade. MLP achieves reasonable performance; classification into pass/fail is practical for early warning systems.

Future work:

- Use more robust architectures (transfer learning is irrelevant here, but ensemble methods or calibration can help).
- Time-series / longitudinal analysis if more temporal data available.
- Add socio-economic features or teacher metrics.
- Use explainability tools (SHAP/LIME) for per-student explanations.

7. References (APA style)

- Cortez, P., & Silva, A. (2008). Using Data Mining to Predict Secondary School Student Performance. *UCI Machine Learning Repository — Student Performance*. [UCI Machine Learning Repository](#)
- Kaggle: Student Performance dataset mirrors. [Kaggle](#)
- Scikit-learn documentation. (Use relevant version.)
- TensorFlow / Keras documentation.

9. Appendix — Sample Code

```
import os  
  
import numpy as np  
  
import pandas as pd
```

```

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.metrics import (
    mean_squared_error, r2_score, confusion_matrix, classification_report,
    roc_curve, auc, accuracy_score)

import tensorflow as tf

from tensorflow.keras import layers, models, callbacks, optimizers

mat = pd.read_csv('student-mat.csv', sep=';')
por = pd.read_csv('student-por.csv', sep=';')
df = pd.concat([mat.assign(subject='math'), por.assign(subject='por')],
               ignore_index=True)

print("Shape:", df.shape)
print(df.info())
print(df.describe(include='all').T)

df['pass_fail'] = (df['G3'] >= 10).astype(int)

plt.figure(figsize=(8,4))

sns.histplot(df['G3'], bins=21, kde=True)

plt.title('Distribution of num_cols =
df.select_dtypes(include=['int64','float64']).columns.tolist()

plt.figure(figsize=(12,10))

sns.heatmap(df[num_cols].corr(), annot=True, fmt=".2f", cmap='coolwarm')

plt.title('Correlation heatmap (numerical features)')

plt.show()

Final Grade (G3)')

plt.xlabel('G3 (final grade)')

plt.ylabel('Count')

plt.show()

plt.figure(figsize=(8,5))

sns.boxplot(x='studytime', y='G3', data=df)

plt.title('G3 by weekly study time (1:<2h, 2:2-5h, 3:5-10h, 4:>10h)')

```

```
plt.show()
plt.figure(figsize=(7,5))
sns.countplot(x='sex', hue='pass_fail', data=df)
plt.title('Pass/Fail counts by sex')
plt.show()
```